

Every Dot Matters: Understanding Inclusivity through Kolam (Rangoli) as an Indian Knowledge Systems Framework

Dr. Savitha G R

Professor – HR, Prin. L N Welingkar Institute of Management Development and Research (WeSchool),
Bangalore

Abstract

Inclusivity is often discussed in organizational and educational contexts through abstract constructs such as belonging, voice, psychological safety, and recognition of uniqueness. This paper proposes that Kolam (a South Indian threshold art drawn around a grid of dots) offers a culturally grounded, practice-based metaphor and analytical framework for understanding inclusivity: every dot has a place, the pattern emerges through relational lines, and the design remains open to iteration without erasing the original grid. This paper positions Kolam—daily threshold drawings constructed around a frame of dots—as an indigenous practice that can illuminate the micro-foundations of inclusion. Kolam’s language rules are simple: dots are placed first; lines then move around them, connecting and enclosing without violating the dot’s integrity. This makes Kolam an unusually powerful metaphor for inclusive systems: every dot matters, and the collective pattern depends on careful recognition of each unit.

Through this research paper attempts to advance the Kolam Inclusivity Model (KIM) with five interlinked dimensions: (1) recognition (dots as persons), (2) relational connection (lines as interdependence), (3) bounded openness (rules that enable participation), (4) symmetry-with-variation (fairness while honoring uniqueness), and (5) renewal (daily re-making as restorative inclusion).

Research questions addressed in this paper are: (RQ1) How can Kolam, as an Indian Knowledge Systems practice, be used to conceptualize and study inclusivity in organizations and management education? (RQ2) What dimensions of inclusion become visible when inclusion is viewed through dot-grids, lines, symmetry, and renewal?

This conceptual paper is based on a qualitative, interpretive methodology combining semi-structured interviews with people who draw Kolam every day, visual elicitation using participants’ designs, and reflexive thematic analysis. The contribution is threefold: a culturally resonant framework to conceptualize inclusion, a methodological template for studying inclusion through everyday arts, and implications for inclusive leadership, DEI pedagogy, and Indian Knowledge Systems (IKS) integration in management education aligned with NEP 2020.

Keywords: inclusivity; belonging; Indian Knowledge Systems; Kolam; Rangoli; arts-based research; visual ethnography; management education.

1. Introduction

Inclusivity has emerged as a central concern across social, educational, and organizational contexts, ref-

lecting a growing recognition that diversity alone is insufficient without genuine participation, belonging, and recognition. Contemporary discourses on inclusivity often rely on policy frameworks, institutional mechanisms, and quantitative indicators. While these approaches are important, they frequently overlook culturally rooted practices that have long embodied inclusive values in subtle, lived, and relational ways. One such practice is Kolam, a traditional South Indian floor art form created using rice flour, chalk powder, or natural pigments, typically drawn at the threshold of homes. Far from being merely decorative, Kolam represents a deeply symbolic, process-oriented, and participatory tradition that offers rich insights into indigenous understandings of inclusivity.

Kolam is traditionally constructed around a grid of dots, with continuous lines drawn to connect, surround, or interlace these dots according to specific rules. The defining feature of Kolam lies not in the final visual outcome alone, but in the disciplined process of ensuring that every dot is acknowledged and integrated into the design. No dot is ignored, erased, or considered insignificant. This foundational principle makes Kolam a powerful cultural metaphor for inclusivity—where each element, regardless of position or prominence, contributes to the harmony and completeness of the whole. In an era where inclusivity is often reduced to symbolic gestures, Kolam offers a practice-based philosophy in which inclusion is both intentional and unavoidable.

Unlike many art forms that privilege symmetry through dominance of central motifs, Kolam distributes attention evenly across the entire pattern. The artist must remain mindful of all dots simultaneously, maintaining balance, continuity, and coherence. This mirrors inclusive systems—whether social, educational, or organizational—where sustainable functioning depends on acknowledging interdependence rather than hierarchy. The absence of a single focal point in many Kolam designs reinforces the idea that collective beauty emerges from relational balance rather than individual prominence. In this sense, Kolam challenges dominant, top-down models of inclusion and instead emphasizes relational inclusion grounded in coexistence.

Kolam is also inherently accessible and participatory. Traditionally practiced by women across caste, class, and educational backgrounds, it does not require formal training, expensive materials, or exclusive spaces. Rice flour Kolams, in particular, serve a dual purpose: aesthetic expression and ecological care, as they feed insects, birds, and small creatures. This act symbolically extends inclusivity beyond human communities to the broader ecological system, aligning with contemporary notions of sustainability and inter-species ethics. Inclusivity, as reflected through Kolam, is therefore not confined to social categories alone but expands into environmental consciousness and everyday ethical practice.

Another significant dimension of Kolam lies in its ephemeral nature. Kolams are drawn daily and erased by wind, footsteps, or rain, only to be recreated the next day. This cyclical process reflects an understanding of inclusivity as a continuous effort rather than a fixed achievement. Inclusion is not permanent or guaranteed; it requires daily attention, care, and renewal. Such a perspective contrasts sharply with institutional approaches that treat inclusivity as a one-time intervention or compliance requirement. Kolam teaches that inclusion is sustained through practice, mindfulness, and repetition.

In contemporary contexts marked by social fragmentation, exclusionary structures, and widening inequalities, revisiting indigenous cultural practices like Kolam provides alternative epistemologies for understanding inclusion. Rather than positioning inclusivity as an imported or externally imposed concept, Kolam demonstrates that inclusive thinking has long existed within local knowledge systems, embedded in routine practices and community life. This perspective is particularly relevant for education, leadership,

and organizational studies, where there is a growing call to integrate indigenous knowledge systems into mainstream theory and practice.

This paper conceptualizes Kolam as a visual–cultural framework of inclusivity, using the metaphor of “every dot matters” to explore how traditional practices can inform contemporary inclusion discourse. By examining the philosophical principles underlying Kolam—interconnectedness, balance, accessibility, and continuity—the study seeks to bridge indigenous wisdom with modern inclusivity frameworks. In doing so, it contributes to a culturally grounded understanding of inclusion that moves beyond rhetoric toward relational, ethical, and practice-based approaches.

2. Literature Review

Across organizations and classrooms, inclusivity is frequently framed as a goal—yet it remains difficult to operationalize beyond policy statements and training modules. Indian Knowledge Systems are conceptualized as a dynamic and practice-oriented body of wisdom encompassing philosophical thought (Savitha, 2025). Contemporary studies increasingly emphasize that inclusion is experienced as both belongingness and being valued for uniqueness, and that these experiences are socially produced through everyday interactions and practices (Shore & Chung, 2023; Li, Ling, & Zhu, 2024). In parallel, arts-based and visual methods have gained prominence for surfacing tacit meaning, embodiment, and context in social life (Nathan et al., 2023; Moralli, 2024). Recent research continues to refine inclusion as a multi-level construct linking organizational practices to individual experiences of belonging. A social network perspective highlights how inclusion is shaped not only by climate but also by meso-level relational structures that enable or constrain access, information, and voice (Workplace inclusion: A social network perspective, 2025).

Leader behaviors remain central. Shore and Chung (2023) argue that leader inclusion can encourage inclusionary behavior while also preventing social exclusion within work groups. Meta-analytic evidence consolidates inclusive leadership’s positive relationships with performance, organizational citizenship behaviors, creativity and voice, while identifying psychological safety, empowerment, perceived inclusion, organizational identification, and leader–member exchange as key mediators (Li et al., 2024). Belonging has also been advanced as a measurable construct. Blau, Goldberg, and Kyser (2023) propose and validate an organizational belonging scale, suggesting that belonging includes acceptance, connection, and the ability to be oneself in role. Related work in higher education shows belonging as a critical driver of engagement and persistence, with systematic reviews consolidating determinants and interventions (Allena et al., 2024; Zaman, 2025).

A conservation-of-resources framing suggests that inclusion processes can be understood through access to resources—material, social, and psychological—especially for under-resourced employees (Fujimoto, Ferdous, & Wali, 2023). This perspective aligns with Kolam’s emphasis on enabling participation through structured supports: the grid and rules are not restrictions, but resource scaffolds that make complex patterns possible. Empirical studies continue to connect inclusive leadership to employee well-being through motivational and relational mechanisms. For example, Liu, Fang, and He (2025) show that inclusive leadership is positively associated with employee well-being, with thriving at work mediating this relationship and growth-need strength acting as a boundary condition. Such findings resonate with Kolam’s ‘renewal’ logic—sustaining inclusion is not a one-off act but a continuous practice of support, correction, and care.

Kolam has received renewed scholarly attention across cultural evolution, design, computation, and heritage science. Tran, Waring, Atmaca, and Beheim (2021) analyze a large corpus of Tamil Kolam and report entropy trade-offs in artistic design, demonstrating how rule-based constraints can still yield rich variation. Sarin (2022) formalizes Kolam as a point lattice system and discusses applications to graphic design, foregrounding the generative role of dot-grids. In HCI and computational creativity, Kannabiran and Reddy (2022) present Kolam as an ecofeminist computational art practice, linking everyday threshold art to sustainability and feminist knowledges. In cultural heritage AI, Sasithradevi et al. (2024) introduce KolamNetV2 for recognizing and classifying Kolam images, reflecting growing interest in digitizing and analyzing Kolam patterns.

Together, these studies support the claim that Kolam is both a cultural practice and a patterned knowledge system—appropriate for theorizing inclusion as structured openness and relational composition.

Arts-based methodologies are increasingly used to explore complex social phenomena involving emotion, embodiment, and tacit knowledge. A systematic narrative review by Nathan et al. (2023) synthesizes arts-based methods and highlights their value in participatory inquiry. Moralli (2024) discusses opportunities and limits of arts-based methods, emphasizing reflexivity and ethical sensitivity. Visual ethnography, including image-based elicitation and analysis, is recognized as generative for qualitative research (Dunlop, 2022), and recent work highlights the need to attend to ethics in visual production and representation (Shortt, 2025).

NEP 2020 calls for curricula rooted in Indian ethos and encourages the integration of Indian knowledge into higher education (Government of India, 2020). The Ministry of Education further articulates IKS as encompassing Jnan, Vignan, and Jeevan Darshan, and provides guidance for incorporation into higher education curricula (Ministry of Education, 2023a, 2023b). These directions legitimize culturally grounded inclusion frameworks such as Kolam for management education, particularly where inclusion, ethics, and sustainability are central outcomes.

3. Research gap

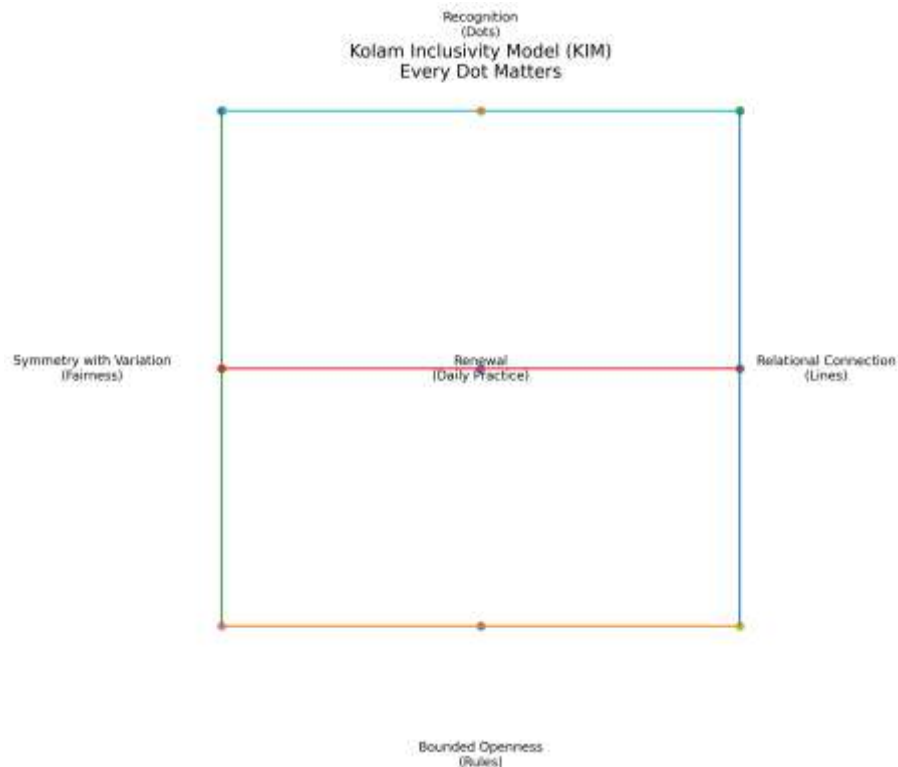
While inclusion research is conceptually rich, it often remains detached from culturally embedded practices that ordinary people use to enact hospitality, recognition, and fairness. Kolam knowledge, in turn, rarely connects the dot-framework practice to inclusion constructs such as belongingness, voice, and psychological safety. This paper bridges these literatures by proposing Kolam as an IKS-derived model for theorizing and studying inclusivity.

4. Conceptual Framework: Kolam Inclusivity Model (KIM)

This paper propose the Kolam Inclusivity Model (KIM), which maps Kolam design principles onto inclusion dynamics. The model comprises five dimensions:

1. Recognition (Dots): Each person is explicitly placed and acknowledged.
2. Relational Connection (Lines): Inclusion emerges through respectful, enabling connections.
3. Bounded Openness (Rules): Clear norms that prevent harm while enabling participation.
4. Symmetry-with-Variation (Fairness + uniqueness): Equity is pursued without forcing sameness.
5. Renewal (Daily practice): Inclusion is sustained through iterative correction and care.

Figure 1 depicting KIM as a dot-framework with five labeled pathways, illustrating how inclusive systems develop from recognition to connection, fairness, and renewal.



5. Methodology

Research design

This study adopts a qualitative, interpretive design combining semi-structured interviews and visual elicitation. The approach is informed by arts-based and visual/ethnographic methods suitable for capturing tacit, embodied practices (Nathan et al., 2023; Dunlop, 2022).

Sampling and participants

Purposive sampling: (a) Kolam practitioners (diverse ages/regions), (b) educators integrating IKS in curriculum, and (c) organizational leaders/HR professionals familiar with inclusion initiatives. A target of 15 interviews supports thematic saturation.

Data collection procedure

Interviews (45–60 minutes) conducted in person and via video call. Participants shared (i) a Kolam they have drawn (or a photo), and (ii) a short narration of design choices. Visual elicitation supported deeper recall and significance; reflexive memos were meticulously recorded.

Data analysis and trustworthiness

Data analyzed using reflexive thematic analysis, iterating between inductive codes (participant meanings) and deductive codes aligned to KIM. Trustworthiness was enhanced through member-checking of summaries, peer debriefing, and through audit trail of coding decisions. Visual artifacts were analyzed for motifs (symmetry, density, boundaries) and interpreted alongside narratives.

6. Discussion and Propositions

Viewing inclusion through Kolam foregrounds a practical insight: inclusion is designed. First, recognition must be explicit—dots are placed before lines are drawn. Second, inclusion is relational—lines create pathways that allow the pattern to exist without erasing individuality. Third, constraints are enabling—

rules prevent exclusion and chaos, similar to how inclusive climates and leader behaviors structure participation (Shore & Chung, 2023).

The research paper three propositions for future testing: (P1) Inclusion increases when recognition practices make each member ‘visible’ before collaborative work begins; (P2) Bounded openness (clear norms + psychological safety) strengthens belonging and voice; (P3) Renewal practices (repair, feedback, second chances) mediate the inclusion–well-being relationship.

7. Implications for Management Education and DEI Practice

For management education, KIM provides a culturally resonant teaching case for DEI concepts, aligning with NEP 2020’s call for education rooted in Indian ethos (Government of India, 2020). Faculty can use Kolam-based activities to teach inclusion as systems design: students map ‘dots’ (stakeholders), ‘lines’ (relationships), and ‘rules’ (norms) to organizational scenarios. For organizations, KIM suggests that inclusion interventions should be assessed not only via climate surveys but also via micro-practices that make people visible, connected, and included (Fujimoto et al., 2023; Li et al., 2024).

8. Limitations and Future Research

This paper advances a culturally grounded conceptual model, but empirical validation remains necessary. Future work can (i) operationalize KIM dimensions and develop measures, (ii) test model pathways across settings, and (iii) compare Kolam-informed inclusion pedagogy with conventional DEI modules on outcomes such as belonging and voice.

9. Conclusion

Kolam demonstrates that inclusivity can be understood as a disciplined aesthetics of togetherness: every dot matters, lines connect without domination, symmetry supports fairness without sameness, and renewal sustains belonging through time. By translating Kolam’s design grammar into the Kolam Inclusivity Model, this paper contributes an IKS-aligned framework for organizational inclusion research and inclusive management education.

References:

1. Allena, K.-A., Slaten, C., Hong, S., Lan, M., Craig, H., May, F., & Counted, V. (2024). Belonging in higher education: A twenty-year systematic review. *Journal of University Teaching & Learning Practice*. <https://open-publishing.org/journals/index.php/jutlp/article/view/767>
2. Blau, G., Goldberg, D., & Kyser, D. (2023). Organizational belonging – proposing a new scale and its relationship to demographic, organization, and outcome variables. *Journal of Workplace Behavioral Health*. <https://doi.org/10.1080/15555240.2023.2178448>
3. Dunlop, S. (2022). Visual ethnography. In *The Wiley Blackwell Companion to Practical Theology* (Chapter 40). Wiley. <https://doi.org/10.1002/9781119756927.ch40>
4. Fujimoto, Y., Ferdous, A., & Wali, F. (2023). Integrative resource model of workplace inclusion for reduced inequality: Conservation of resources perspective. *Journal of Business Ethics*, 187(2). <https://doi.org/10.1007/s10551-022-05218-7>
5. Government of India. (2020). National Education Policy 2020. Ministry of Education. https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English_0.pdf

6. Kannabiran, G., & Reddy, A. V. (2022). Exploring Kolam as an ecofeminist computational art practice. In Proceedings of the 14th Conference on Creativity and Cognition (C&C '22) (pp. 336–349). Association for Computing Machinery. <https://doi.org/10.1145/3527927.3531452>
7. Li, X., Ling, C.-D., & Zhu, J. (2024). Implications of inclusive leadership for individual employee outcomes: A meta-analytic investigation of the mediating mechanisms and boundary conditions. *Asia Pacific Journal of Management*. <https://doi.org/10.1007/s10490-024-09987-9>
8. Liu, Y., Fang, Y., & He, G. (2025). The effect of inclusive leadership on employee wellbeing. *Scientific Reports*, 15, 39991. <https://doi.org/10.1038/s41598-025-23703-7>
9. Ministry of Education. (2023a). Indian knowledge systems. Government of India. <https://www.education.gov.in/en/nep/indian-knowledge-systems>
10. Ministry of Education. (2023b). Guidelines for incorporating Indian knowledge in higher education curricula. Government of India. <https://www.education.gov.in/en/guidelines-incorporating-indian-knowledge-higher-education-curricula>
11. Moralli, M. (2024). Arts-based methods in migration research: A reflection on potentialities and limits. *International Journal of Qualitative Methods*. <https://doi.org/10.1177/16094069241254008>
12. Nathan, S., et al. (2023). The use of arts-based methodologies and methods with young people with complex psychosocial needs: A systematic narrative review. *Health Expectations*, 26. <https://doi.org/10.1111/hex.13705>
13. Sarin, A. (2022). The Kolam drawing: A point lattice system. *Design Issues*, 38(3), 34–54. https://doi.org/10.1162/desi_a_00690
14. Sasithradevi, A., et al. (2024). KolamNetV2: Efficient attention-based deep learning model for Tamil kolam classification. *Heritage Science*. <https://www.nature.com/articles/s40494-024-01167-8>
15. Savitha, (2025), Integrating Indian knowledge system (IKS) into Management Education: A Conceptual Framework, *International Journal of Innovative Research in Technology - An UGC-Compliant International Research Journal*, Volume: 12, Issue: 7, Page No: 4919-4922
16. Shore, L. M., & Chung, B. G. (2023). Enhancing leader inclusion while preventing social exclusion in the work group. *Human Resource Management Review*, 33(1), 100902. <https://doi.org/10.1016/j.hrmr.2022.100902>
17. Shortt, H. (2025). Visual methods and research philosophy. In *Visual Methods and Research Philosophy* (Chapter). Edward Elgar. <https://www.elgaronline.com/abstract/book/9781839106835/chapter2.xml>
18. Tran, N.-H., Waring, T., Atmaca, S., & Beheim, B. A. (2021). Entropy trade-offs in artistic design: A case study of Tamil kolam. *Evolutionary Human Sciences*, 3, e23. <https://doi.org/10.1017/ehs.2021.14>
19. Workplace inclusion: A social network perspective. (2025). *Organizational Behavior and Human Decision Processes*. <https://www.sciencedirect.com/science/article/pii/S0191308525000036>
20. Zaman, M. (2025). Students' sense of belonging in higher education: A systematic review of antecedents and interventions. *Teaching in Higher Education*. <https://doi.org/10.1080/13562517.2024.2406069>

Exhibit 1

Interview protocol (sample questions)

- When you draw Kolam, what does placing the dots mean to you? Who or what is being 'accounted for' in the dot grid?

- How do you decide which dots are connected, and which paths the lines will follow? What happens if a dot is missed?
- What ‘rules’ do you follow in Kolam, and why are those rules important? What happens when rules are broken?
- Do you see any connection between Kolam and welcoming others or creating a sense of belonging at home/community?
- Where do you see ‘fairness’ in Kolam—symmetry, spacing, balance? How do you handle variation and creativity?
- Can you recall a time someone learned Kolam from you (or you learned from someone)? What made the learning inclusive or exclusive?
- If Kolam were used to explain inclusion at work/college, which parts translate well and which do not?
- What does it mean to re-draw Kolam daily? Does it relate to repair, second chances, or restoring harmony?